Ontology Unpacking and Semantic Bridging for Enterprise Decision-Making

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Abstract. Enterprise decision-making frequently depends on two semantically disconnected forms of organizational knowledge: fragmented business evidence and underspecified business intent. Although recent AI approaches can retrieve relevant correlations and generate contextual suggestions, they often lack an explicit semantic mechanism for determining whether the available evidence formally satisfies the intended decision conditions. This paper proposes an ontology-guided methodology for enterprise decision-making structured around three integrated stages: Semantic Onboarding, Ontological Unpacking, and Semantic Bridging. The approach transforms distributed transactional evidence and high-level business criteria into ontologically grounded formal artifacts and compiles them into a unified executable inference structure. By explicitly aligning business truthmakers with formalized decision intent, the framework enables decision outcomes to be generated through logical satisfiability rather than probabilistic association alone, guided by the Unified Foundational Ontology (UFO) and materialized as executable Prolog knowledge models. We instantiate the methodology in strategic sales recommendation tasks, including discount prioritization, bundle construction, and personalized offer generation. Comparative evaluation against a Knowledge Graph- and Large Language Model-based recommender baseline shows consistently superior semantic coherence and explainability. At the same time, scalability experiments demonstrate stable millisecond-level inference under substantial growth in the fact base.

Keywords: Decision-Making, Explainable AI, Ontology-Driven Systems, Semantic Bridging, Ontological Unpacking, Logic-Based Reasoning.

1 Introduction

Enterprise decision-making often depends on the ability to reconcile two distinct forms of organizational knowledge. On one hand, relevant evidence is usually fragmented across distributed business records such as transactional histories, product metadata, operational indicators, and descriptive organizational attributes. On the other hand, the decision objective itself is rarely explicitly formalized, remaining expressed only through high-level strategic intentions, tacit

expert assumptions, or loosely stated business criteria. As a result, organizations frequently possess both the raw evidence and the intended goal but lack an explicit semantic mechanism to determine whether the available evidence truly satisfies the intended decision conditions.

Recent advances in Artificial Intelligence have sought to mitigate this gap by integrating Large Language Models (LLMs) with Knowledge Graphs (KGs). These approaches are often capable of retrieving relevant relational patterns, identifying contextual associations, and generating natural-language explanations over structured data. However, their reasoning process remains predominantly correlation-driven: entities are connected via weighted edges, embeddings, or semantic similarity signals, while the final decision is typically inferred from probabilistic associations rather than explicit semantic verification. Consequently, such systems can explain why entities appear related in the data. However, they still struggle to formally establish why a target organizational decision should hold, a limitation that becomes critical in scenarios where traceability, justification, and semantic validity are required.

To address this limitation, we propose an ontology-driven methodology for enterprise decision-making that combines ontological unpacking, semantic alignment, and logic-based reasoning. Rather than relying solely on observed data associations, the approach transforms both transactional evidence and explicit business knowledge into a semantically grounded executable knowledge base capable of reasoning about higher-level organizational objectives. As a running example, we instantiate this methodology in strategic sales recommendation tasks such as discount prioritization, bundle suitability, and alternative sales opportunities. In the illustrative scenario considered throughout this paper, products are associated with brands and categories, customers perform purchase transactions, and recommendation decisions are derived from both transactional evidence and business-defined promotional strategies. The objective is to determine which products, categories, or product combinations should be prioritized in marketing campaigns. By representing business relations as ontological truthmakers and aligning them with onboarded organizational evidence, the framework supports the generation of deductive decisions. It produces explainable outputs whose strategic justifications are fully traceable. In this context, while KG-based recommender systems explain product correlations, the proposed methodology explains why specific enterprise actions hold.

The remainder of this paper is organized as follows. Section 2 presents the ontological and neuro-symbolic foundations underlying the proposed approach. Section 3 reviews the main related research streams in semantic enterprise decision support. Section 4 introduces the proposed three-stage methodology, while Section 5 details its architecture and implementation. Section 6 reports the comparative experimental results and scalability assessment. Finally, Section 7 concludes the paper and outlines directions for future work.

2 Background

Contemporary decision-support systems increasingly rely on neuro-symbolic architectures that seek to combine the representational flexibility of neural models with the traceability and deductive rigor of symbolic reasoning. According to [22], Neuro-Symbolic Artificial Intelligence (NeSy AI) emerges from integrating neural and symbolic traditions to combine distinct yet complementary forms of computational representation. While neural systems like Large Language Models (LLMs) encode information through distributed patterns of weighted activations that are highly effective for learning from raw data, such representations are largely opaque to human observers. Symbolic systems, in contrast, represent knowledge explicitly through manipulable symbols, rules, and formal structures that are inherently interpretable, logically verifiable, and closely aligned with human reasoning. According to [23], traditional neural networks excel at pattern recognition and data-driven learning but lack interpretability and logical reasoning capabilities. Conversely, symbolic reasoning provides structured knowledge representation and rule-based inference but struggles with handling large-scale, unstructured data.

In organizational environments, this combination is particularly relevant because decision propositions often depend simultaneously on heterogeneous enterprise evidence and on formal business constraints that cannot be reliably inferred from statistical patterns alone [10]. As a consequence, the construction of executable symbolic knowledge becomes a central requirement for semantically robust decision support. Traditional enterprise data integration pipelines partially address this problem by consolidating heterogeneous sources into centralized repositories. However, the resulting artifacts are usually optimized for storage and retrieval rather than for semantic executability. Recent works on knowledge engineering emphasize the integration of heterogeneous information, rich ontologies, and semantics towards knowledge-aware models [13].

The Unified Foundational Ontology (UFO) was developed to provide a rigorous ontological basis for conceptual modeling and information systems engineering. Unlike lightweight taxonomies, UFO offers richer distinctions among entities, relations, events, and intentional structures. UFO is organized into three complementary modules: UFO-A, devoted to structural entities that persist in time; UFO-B, devoted to temporally unfolding events and processes; and UFO-C, devoted to social and intentional aspects of reality [9].

Within UFO-A, a central concern is the principle of identity of domain entities, a view influenced by OntoClean and by the philosophical contributions of [24] and [18]. In this layer, *Kinds* provide rigid identity criteria for their instances. At the same time, UFO distinguishes substantials as *Functional Complexes*, i.e., functionally organized individuals, and *Collectives*, i.e., individuals composed of homogeneous members [7]. UFO-A also introduces the distinction between *internal relations*, which derive from intrinsic properties, and *material relations*, which require a mediating ontological entity called a *relator* [4]. Relators ground the existence of externally valid relations and are existentially dependent on the entities they connect, a dependence captured by the notion of *foundation* [4].

Table 1. Conceptual mapping between UFO constructs and their executable counterparts in the proposed methodology.

UFO Construct	Executable Representation
Agent	`agent(User,...)`
Functional Complex	`functional_complex(Product,...)`
Relator	`unpack_relation(...)`
Material Relation	`semantic_recommendation(...)`

UFO-B complements this structural perspective by modeling *perdurants*, i.e., events and processes that unfold temporally [9]. This temporal view is particularly relevant because many material relations emerge through events, such as purchase situations that instantiate user-product commercial interactions. Such ontological grounding naturally connects to the notion of *truthmakers*: according to [3], a truthmaker is an entity whose existence makes a proposition true. In this sense, relators function as truthmakers for material business relations. Finally, UFO-C introduces *Agents* and *Roles*, enabling entities to participate in intentional, socially contextualized situations [6]. Together, these three layers provide the ontological basis for treating recommendation evidence not as simple observed associations, but as semantically grounded commercial facts.

Table 1 summarizes how the main UFO constructs employed in this work are operationalized within the executable semantic model. Agents and functional complexes correspond to domain entities identified during Semantic Onboarding, while relators and material relations emerge during Ontological Unpacking as semantic abstractions over observable business associations.

3 Related Work

Recent advances in enterprise decision systems have increasingly highlighted the limitations of conventional AI architectures for semantically grounded decision support. While many modern systems are capable of aggregating organizational records, retrieving contextual information, and generating predictive suggestions, they often rely on surface-level correlations rather than on explicit semantic verification that the available evidence truly satisfies an intended organizational objective. This limitation has motivated a broader movement toward enterprise knowledge representations that preserve not only data accessibility, but also machine-interpretable semantic commitments.

Within this movement, Knowledge Graph research has become one of the dominant paradigms for enriching organizational information with explicit entities, typed relations, schema constraints, and contextual semantics. According to [11], this represents an important transition from simple data interoperability to semantic interoperability. Nevertheless, as surveyed in [13], most Knowledge Graph construction approaches remain primarily concerned with acquisition, completion, and representational enrichment, leaving open the question of how such semantic artifacts can be systematically transformed into executable inferential models that support decision-oriented organizational reasoning.

This unresolved transition from semantic representation to semantic execution naturally motivates both ontology-based conceptual modeling and neuro-symbolic artificial intelligence. Foundational ontologies provide semantic precision beyond conventional information schemas by explicitly representing the roles, events, dependencies, and relational commitments that underlie organizational phenomena [9]. In parallel, neuro-symbolic AI seeks to combine the adaptive learning capabilities of neural models with the transparency, traceability, and formal reasoning strengths of symbolic systems [2, 15]. Together, these approaches improve conceptual fidelity, explainability, and relational reasoning. However, despite these advances, ontology-based modeling has remained largely focused on conceptual consistency and model validation. At the same time, most neuro-symbolic approaches assume that symbolic artifacts already exist and devote comparatively little attention to the methodological challenge of constructing executable knowledge structures from heterogeneous organizational evidence and partially specified business objectives.

An additional unresolved issue concerns the formalization of decision intent itself. Organizational decisions rarely depend on business evidence alone; they also require semantic validity criteria, contextual thresholds, and evaluative constraints that define when a conclusion becomes decision-relevant. Such conditions are frequently only partially declared across policies, expert narratives, and tacit institutional practices. Research in goal-oriented requirements engineering has long recognized that human objectives and evaluative conditions must undergo explicit refinement before they can sustain computable reasoning [16]. However, most existing decision-support architectures assume these decision semantics as pre-modeled formal inputs, rather than addressing how partially specified decision intent can itself be transformed into machine-executable semantic structures and then systematically connected to organizational evidence.

These research streams reveal a methodological fragmentation. KG research improves semantic representation, ontology-based modeling improves conceptual precision, neuro-symbolic AI improves explainable symbolic reasoning, and requirements engineering improves the formal treatment of evaluative objectives. What remains insufficiently addressed is a unified semantic construction pipeline capable of jointly transforming fragmented enterprise evidence and partially specified decision intent into a single executable inferential structure.

4 Methodology

The proposed methodology establishes a formal pipeline to bridge the gap between business intent and organizational knowledge to support decision-making. As illustrated in Figure 1, our methodology is structured as a multi-stage pipeline designed to transition from raw information to ontologically unpacked knowledge. The pipeline is divided into three self-contained stages: Semantic Onboarding, Ontological Unpacking, and Semantic Bridging. It begins with two distinct inputs: *raw data*, such as transactional logs in CSV or RDF formats, and *business knowledge*, tacit expertise from domain specialists. These inputs undergo a

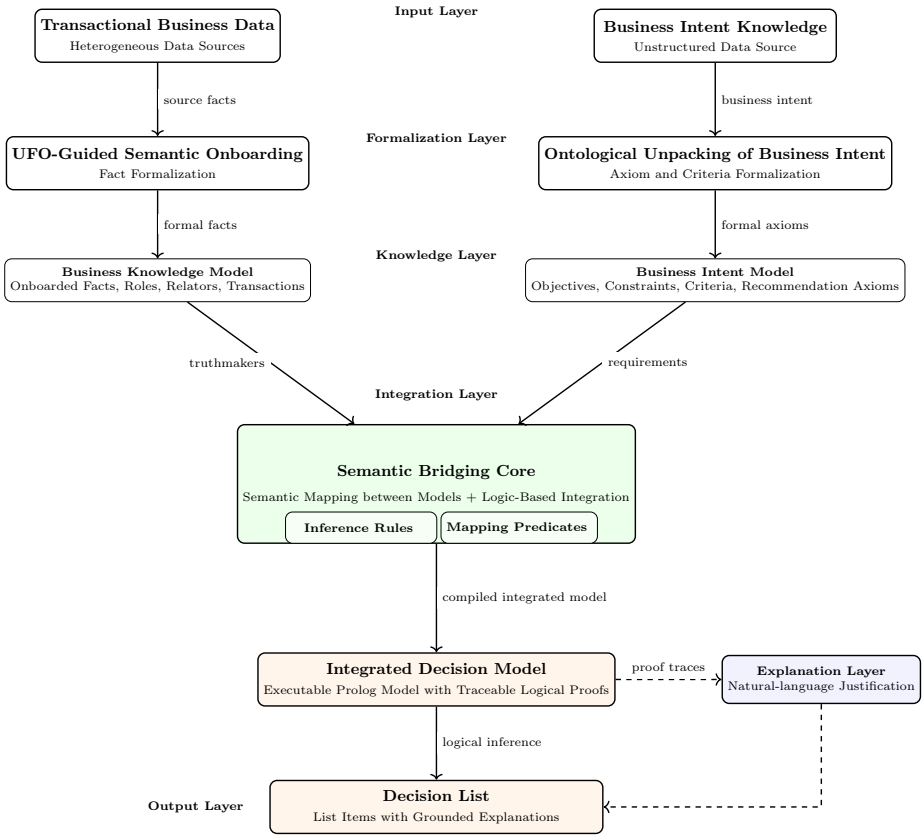


Fig. 1. Semantic Bridging methodology for integrating business knowledge and business intent into an executable decision model.

formalization process through *Semantic Onboarding* and *Ontological Unpacking*, respectively. While the former focuses on instantiating entities within a foundational ontology, the latter codifies the "rules of the world" that govern how relationships should be interpreted.

The integration of these flows occurs in the *Semantic Bridging* stage. This layer acts as the execution engine where abstract axioms meet concrete instances. By unifying these semantic spaces, the framework moves beyond simple pattern matching, identifying the specific *relators* and *truthmakers* that justify an action; the process culminates in a decision list, where the logical proof generated during the bridging phase serves as the ground truth for the final natural language explanation, effectively mitigating hallucinations and ensuring high-fidelity reasoning.

4.1 UFO-Guided Semantic Onboarding of Raw Data

This stage focuses on the formal ingestion of raw dataset entities. Through Semantic Onboarding, we perform a semantic lift of the data, transforming flat records into a formal Knowledge Base (KB). Following the principles of the Unified Foundational Ontology (UFO), each entity is categorized into foundational types. This ensures that the system recognizes the data's existential nature, distinguishing between *agents* (users with intentionality) and *functional complexes* (artifacts with functional roles).

To illustrate this process, we introduce a running example centered on a retail organization preparing a strategic promotional sales campaign. The organization's objective is to maximize overall profitability by identifying (i) product pairs suitable for bundle discounts, (ii) product categories that should be prioritized for discount incentives, and (iii) personalized product recommendations for specific customers. In this case, the system must reason about explicit business sales criteria to determine which promotional actions are most likely to generate commercial gain. In this example, a raw record $\{id : U0534, name : JamesWeber\}$ is transformed into the fact $agent('U0534', 'JamesWeber', \ldots)$. A product $P102$ is mapped as $functional_complex('P102', \ldots, 'B028', 'C137')$, grounding the data into explicit ontological categories such as agents, functional complexes, and their associated business classifications.

4.2 Ontological Unpacking of Business Intent

The Ontological Unpacking phase defines the rationale of the target decision by formalizing the ontological validity conditions that must hold for a decision proposition to become semantically admissible. In UFO, a relation between two entities is not treated as a mere observable association, but as a proposition grounded on the existence of a mediating *relator*. Accordingly, this phase decomposes high-level business objectives and opaque domain relations into formal semantic conditions that specify under which organizational circumstances such a relational truthmaker would be considered valid. In the running example, a graph relation such as `also_buy` is semantically unpacked into a *synergy* relator that formally mediates the strategic association between two products. This relator is then supported by explicit business conditions, such as brand affinity, commercial complementarity, or shared campaign suitability, which establish the ontological admissibility criteria for the resulting recommendation within the business domain.

A key distinction of this methodology is that the axiomatic foundations of Ontological Unpacking are derived from high-level natural language business specifications. Typically provided by domain experts or business managers, these descriptions encode organizational concepts, category definitions, and strategic constraints. The Unpacking process employs an LLM to identify candidate relators, roles, and business constraints, translating this qualitative knowledge into formal logical rules aligned with the domain's semantic nuances. The generated constructs are subsequently reviewed by a domain analyst, who verifies their

semantic compatibility with the UFO conceptualization and either accepts, refines, or rejects them before inclusion in the Business Intent Model. Constructs lacking a valid UFO grounding are not incorporated into the executable model. During this validation stage, predicates are also classified as mandatory or optional according to their relevance to the intended decision objective.

This phase also establishes a direct connection between real-world domain knowledge and the later LLM-assisted explanation process. Because business criteria are formalized as ontology-guided logical rules, modifications to the recommendation strategy can be incorporated by updating the natural-language specifications that generate the ruleset, without requiring changes to the reasoning architecture. As a result, domain experts remain responsible for defining the real-world facts that serve as *truthmakers*, ensuring that every decision remains logically grounded and contextually faithful.

The ontological constructs generated during Ontological Unpacking are not used merely as labels. They become explicit predicates that participate directly in recommendation rules and satisfiability checks. Consequently, removing these constructs would alter the set of admissible recommendations and the resulting logical evaluations. This behavior can be directly inspected in the accompanying artifact repository, which includes the generated executable models, logical rules, and evaluation scripts used in the experiments.

4.3 Semantic Bridging

The final stage, Semantic Bridging, integrates the two previously independent formal spaces produced by Semantic Onboarding and Ontological Unpacking. Its role is to allow the abstract business axioms derived from intent to find their corresponding *truthmakers* in the onboarded business facts. Although both layers are individually formalized, they remain disconnected until explicit semantic correspondences are established between compatible ontological components.

This integration is performed by identifying, for each target decision requirement, the set of onboarded entities and relations that satisfy its ontological conditions. In practical terms, abstract decision predicates are transformed into executable predicates whose satisfiability can be tested against concrete organizational evidence. In the running example, predicates such as *bundle_candidate(X,Y)*, *discount_priority(C)*, or *personalized_offer(U,P)* become inferable only when the onboarded transactional facts can instantiate the unpacked *synergy* and business-validity conditions defined in the previous stage. The result of this process is a new formal artifact, called the *Executable Semantic Decision Model*, which consolidates in a single executable layer: (i) the organizational objectives derived from business intent, (ii) the truth-bearing business facts extracted from data, and (iii) the bridging rules that determine under which semantic conditions a target decision proposition becomes valid.

Because the alignment between business predicates and onboarded facts is not assumed to be perfectly complete, the Semantic Bridging engine performs a constrained satisfiability check in which each recommendation rule is decomposed into mandatory and optional conditions. Mandatory predicates must find

valid transactional or ontological truthmakers for the recommendation to be generated, whereas optional predicates provide additional semantic support when available. If a mandatory condition fails, the recommendation candidate is discarded; if only optional evidence is absent, the recommendation may still be produced with lower formal completeness, which is later reflected in the Logical Soundness evaluation.

Finally, because an LLM assists in translating natural-language business strategies into executable Prolog rules, all generated axioms undergo expert-in-the-loop verification before execution. This validation is performed by a domain analyst familiar with both business intent and formal logic, ensuring that the unpacked recommendation rules preserve the intended commercial semantics before being admitted into the Semantic Bridging layer.

5 Architecture and Implementation

The system architecture is organized into functional modules that map to the pipeline stages. It is implemented in Python, leveraging its ecosystem for data manipulation and AI integration, and uses SWI-Prolog as the core symbolic reasoning engine. The architecture emphasizes the separation between the Data Ingestion layer (Python-based), the Knowledge Modeling layer (Natural Language-to-Prolog translation supported by LLMs), and the Reasoning Layer (executed within the Prolog environment), which provides grounded explanations and synthesizes the logical output into a human-readable format.

The Data Ingestion module processes raw data sources using custom Python parsers that transform them into standardized internal representations. These internal representations consist of structured Python objects that normalize heterogeneous source formats before their conversion into executable Prolog predicates. The Semantic Onboarding engine maps parsed data to the Unified Foundational Ontology (UFO). It is responsible for generating two types of Prolog files: entities.pl, which contains typed individual facts, and triples.pl, which stores the material relations. Table 2 illustrates the Semantic Onboarding process, which transforms raw data into structured, semantically meaningful representations. This transformation is a key contribution of the proposed methodology, as it enables the conversion of raw data sources into a unified, logic-based representation suitable for reasoning. By generating Prolog predicates that explicitly encode entities and their relations, the onboarding process establishes the structural foundation required for subsequent semantic bridging and inference.

The Ontological Unpacking Modeler supports translating natural-language business rules into formal Prolog axioms and generating the rules.pl file. We utilize Gemini 2.0 Flash, through a single-shot prompting strategy and the provider's default generation parameters, to codify qualitative descriptions, for example *customers value brand consistency*, into precise logical clauses that identify the corresponding UFO *Relators*. As mentioned in Section 4, all generated rules are reviewed and validated by a domain analyst before being admitted into the executable model. Table 3 presents the Ontological Unpacking process, in

Table 2. Examples of data ingestion and semantic mapping into Prolog.

Stage	Representation
Example 1	

Product ingestion	`product_id, product_name, brand_id, category_id`
	`P102, Samsung 980 SSD, B028, C137`
	`P205, Samsung Odyssey Monitor, B028, C141`

Semantic mapping	`functional_complex('P102','Samsung 980 SSD','B028','C137').` `functional_complex('P205','Samsung Odyssey Monitor','B028','C141').`
Example 2	
Transaction ingestion	`user_id, product_id, timestamp` `U0534, P102, 2023-11-15T10:00:00Z`
Semantic mapping	`buys('U0534','P102','2023-11-15T10:00:00Z').`

Table 3. Example of translating a business strategy into executable Prolog rules.

Stage	Representation
Business Strategy	
Natural-language intent	`Suggest products from the same brand to leverage brand loyalty (Ecosystem Synergy).`
Semantic interpretation	`ecosystem_synergy(P1, P2)`
Executable rule generation	`unpack_relation(P1,P2,ecosystem_synergy) :-` `functional_complex(P1,_,B,_), functional_complex(P2,_,B,_), P1 \= P2.`

which high-level business strategies expressed in natural language are translated into executable logical rules.

The Semantic Bridging process is implemented as a compilation layer of meta-interpreter rules within SWI-Prolog, responsible for integrating the two formal artifacts generated in the previous stages: the *Business Knowledge Model* and the *Business Intent Model*. Rather than querying these models independently, the architecture dynamically loads both sets of predicates. It generates an additional executable inference layer comprising mapping predicates, satisfiability predicates, and decision rules, thereby materializing the *Integrated Strategic Recommendation Model*. This constrained satisfiability check evaluates whether all mandatory logical conditions associated with a recommendation are satisfied within the executable knowledge model. This process is implemented through standard SWI-Prolog rule evaluation and backtracking.

At runtime, the Prolog engine performs deductive unification on this integrated model to determine whether a target decision proposition is logically satisfiable. In practical terms, this means verifying whether the evidence for the onboarded business jointly satisfies the formal axioms derived from the organizational decision intent. As a result, decision outcomes are not produced by heuristic ranking or graph traversal, but by formal proof construction. In the running example, the inference process may establish that two products satisfy the ontological and transactional conditions required for a bundle-discount campaign, or that a given product category meets the strategic criteria for promotional-discount prioritization. Each generated outcome, therefore, emerges as an executable logical conclusion whose supporting *truthmakers* remain explicitly identifiable.

To make these formal proofs accessible to decision makers, the architecture includes an LLM-Assisted Explanation module. This component receives the

structured logical derivation produced by the Prolog engine and translates it into a natural-language justification. Importantly, the LLM is not responsible for generating the decision outcome itself, but only for verbalizing the already validated proof traces. Prompt constraints require the explanation to remain strictly anchored in the logical evidence supplied by the reasoning layer, ensuring that the final textual justification is a faithful semantic rendering of the underlying formal proof.

6 Evaluation

To evaluate the proposed approach, we compare it against a recommender system based on Knowledge Graphs and Large Language Models, as described in [25]. In the baseline architecture, domain knowledge is represented as a Knowledge Graph in which entities are connected through weighted relations such as *also_buy*, *also_view*, and *belongs_to*. These RDF triples are transformed into textual representations and injected into the LLM prompt together with domain-specific instructions that guide recommendation generation and explanation.

6.1 Evaluation Strategy and Metrics

Both our proposed framework and the baseline approach are evaluated against the same underlying business dataset, which contains 5,200 entities and 18,600 relationships, and the same set of three core Competency Questions (CQs). These questions materialize the business intent and act as the benchmark for our reasoning engine, forcing the system to navigate the ontological structure and derive answers from formal rules. These questions comprise CQ1 (Predictive Reasoning: *What other products a user could buy?*), which evaluates the ability to identify items based on user profile and semantic similarity; CQ2 (Strategic Reasoning: *Which product categories should be prioritized for a discount?*), which assesses reasoning over category-level ontologies and business rules; and CQ3 (Relational Reasoning: *Which two products would be the most suitable candidates for a bundle discount?*), which tests the capacity to identify *relator* dependencies between items using UFO-based modeling.

The three competency questions exercise the framework's complementary reasoning capabilities, covering personalized recommendation (CQ1), category-level strategic prioritization (CQ2), and relational bundle recommendation (CQ3). Together, they evaluate the Semantic Bridging architecture's ability to support customer-centered, category-centered, and campaign-oriented decision-making within a unified, executable inference framework.

To evaluate the effectiveness of the proposed framework, we adopted two complementary metrics capturing both formal recommendation validity and practical decision usefulness. The first metric, *Logical Soundness (LS)*, is a quantitative measure defined as the average percentage of mandatory ontological and business axioms that each generated recommendation satisfies for its corresponding competency question. For each recommendation, the Semantic Bridging layer

Table 4. Representative outputs for the three strategic sales competency questions.

CQ	Strategic Objective	Representative Output
CQ1	Personalized recommendation	Apple Ergonomic Monitor suggested as a complementary offer.
Truthmakers & Justification		Previous laptop purchase + ontological Solution Composition relation between acquired and candidate products. The recommendation is justified as a complementary product, based on prior computing acquisitions and semantic compatibility.
CQ2	Category discount prioritization	Category C199 selected for discount campaign.
Truthmakers & Justification		Category mentioned by user + 3 prior purchases + 6 also-buy targets. The recommendation is justified by high historical affinity and strong downstream cross-selling potential.
CQ3	Bundle discount recommendation	Samsung Ergonomic Accessories + Dell Ultrawide Storage identified as the best bundle pair.
Truthmakers & Justification		`also_buy = true` + `also_view = true` + cross-category complementarity. The recommendation is justified as a joint promotional offer supported by strong co-consumption behavior.

verifies how many of the formal predicates required by the intended strategic sales objective, including transactional truthmakers, ontological mapping conditions, and strategic inference constraints, are effectively satisfied during logical execution. Thus, LS does not simply indicate whether a recommendation can be produced, but measures the degree to which the recommendation is logically complete with respect to the full set of formal business conditions expected for that strategic task.

For each recommendation r, let $C_q = \{c_1, c_2, \ldots, c_m\}$ be the set of formal conditions required by competency question q. Each condition corresponds to a Prolog-verifiable axiom, such as the existence of a transactional truthmaker, an ontological mapping predicate, an intent-alignment condition, or a strategic inference constraint. Let satisfied(c_i, r) denote that condition c_i is satisfied by recommendation r. Logical Soundness is defined as:

$$LS(r) = \frac{|\{\, c_i \in C_q \;:\; \text{satisfied}(c_i, r)\,\}|}{|C_q|} \times 100$$

The LS score for each competency question is the average LS across the generated recommendations for that question. This metric is objective because each condition is checked against formal predicates, row-level evidence, or Prolog execution results, while still allowing partially supported recommendations to receive intermediate scores. For reporting purposes, Table 5 includes a *Mean Satisfied Axioms* column, which represents the average number of satisfied axioms divided by the average number of axioms evaluated for recommendations generated under each competency question. This measure provides an interpretable view of the underlying LS computation by explicitly showing the average level of axiom satisfaction achieved by the generated recommendations.

The second metric, Expert Rating (ER), is a qualitative assessment performed by six domain specialists using a 1–5 Likert scale. Evaluators independently assessed all recommendations generated for each competency question using a blind evaluation protocol, evaluating the strategic relevance, commercial

Table 5. Quantitative and qualitative evaluation results across the three competency questions.

CQ	Mean Satisfied Axioms	LS (%)	ER Base.	ER Prop.
CQ1: Personalized Rec.	5.00 / 6.00	83.33	3.4	4.6
CQ2: Category Discount	4.00 / 7.00	57.14	3.1	4.3
CQ3: Bundle Suitability	5.55 / 7.00	79.29	2.9	4.1
Overall Mean	**5.37 / 6.78**	**79.37**	**3.1**	**4.3**

actionability, and explanatory quality of each recommendation. The ER values reported in Table 5 correspond to the mean scores assigned by the evaluators across all recommendations generated for each competency question.

The evaluation results showed a high LS (79.29%), indicating that most bundle-discount competency questions were answered correctly. From a quantitative perspective, the *Logical Soundness* values indicate that the generated recommendations, on average, satisfy a substantial portion of the mandatory ontological and business axioms encoded for each strategic sales task. CQ1 achieved the highest LS (83.33%), showing that personalized recommendation is strongly supported when customer purchase history can be consistently connected to semantic product compatibility relations. CQ3 also showed a high LS (79.29%), indicating that most bundle-discount candidates satisfy the expected combination of co-purchase evidence, relational complementarity, and strategic bundling conditions. In contrast, CQ2 obtained the lowest LS (57.14%), suggesting that category-discount prioritization depends on a broader set of aggregate business criteria, some of which were only partially verifiable by the current formal predicates. Overall, the mean LS of 79.37% indicates that the Semantic Bridging layer generally produces recommendations that are formally close to the intended strategic business conditions.

Regarding the qualitative assessment, the proposed framework received consistently high *Expert Ratings* across the three scenarios, with an overall mean of 4.3, indicating that the generated decision recommendations were perceived as strategically coherent, commercially actionable, and supported by clear justifications. Personalized recommendation again received the highest qualitative evaluation, followed by category discount prioritization and bundle discount suitability. Interestingly, even in CQ2, where Logical Soundness was lower, the expert assessment remained positive, suggesting that evaluators recognized some commercially meaningful business signals despite these not being fully captured by the strict formal verification process. Compared with the baseline ratings, the proposed recommendations were generally perceived as more convincing from a strategic sales perspective, reinforcing that the framework's contribution lies not only in formal semantic grounding but also in producing recommendations that better support practical promotional decision-making.

Table 6. Scalability results of the Semantic Bridging inference engine under increasing fact volumes.

Approx. Facts	Load Time (ms)	CQ1 (ms)	CQ2 (ms)	CQ3 (ms)	Mean Query Time (ms)
8,005	1,291.62	3.71	7.82	5.29	5.61
40,005	6,728.19	4.28	0.89	3.97	3.05
80,005	47,180.15	3.22	1.15	2.94	2.44
200,005	213,887.13	6.35	3.02	4.31	4.56
400,005	1,062,286.39	11.39	7.17	6.22	8.26

6.2 Scalability Assessment

We evaluated the computational scalability of the proposed Semantic Bridging engine by measuring the behavior of the SWI-Prolog inference layer under progressively larger knowledge bases. All scalability experiments were executed using the standard Google Colab environment without dedicated computational resources. Following prior evidence [31] that SWI-Prolog provides a robust and scalable environment for knowledge-intensive applications, we generated five datasets ranging from approximately 8 thousand to 400 thousand onboarded facts while preserving the same recommendation rules and competency queries. For each configuration, we measured both the knowledge base loading time and the average inference time required to execute CQ1, CQ2, and CQ3.

Despite the growth of the fact base, recommendation inference remained highly stable across all evaluated configurations. Even for the largest dataset, containing more than 400 thousand facts, all three competency questions were answered in less than 12 milliseconds on average. These results indicate that the proposed semantic recommendation architecture is computationally viable for large business repositories, with the primary scalability cost concentrated in fact ingestion rather than in logical recommendation generation. Non-monotonic behavior was observed for CQ2, with query execution time decreasing from 7.82 ms at 8K facts to 0.89 ms at 40K facts. This behavior is likely due to optimization mechanisms in the underlying Prolog engine, including indexing and query execution strategies that may become more effective as the fact base grows. Since the reported execution times are in the millisecond range, minor variations due to caching and runtime effects may also affect individual measurements.

7 Conclusions and Future Work

This paper presented an ontology-guided framework for enterprise decision-making to address a critical semantic gap in organizational decision systems: the disconnect between high-level business intent and the business evidence available in operational data. By explicitly formalizing both organizational decision criteria and underlying business facts, the proposed approach enables decision generation to be treated as a formal satisfiability problem rather than an association retrieval task. The Semantic Bridging layer integrates these previously disconnected semantic spaces into a unified executable inference structure, allowing decision outcomes to be produced through traceable logical proof construction.

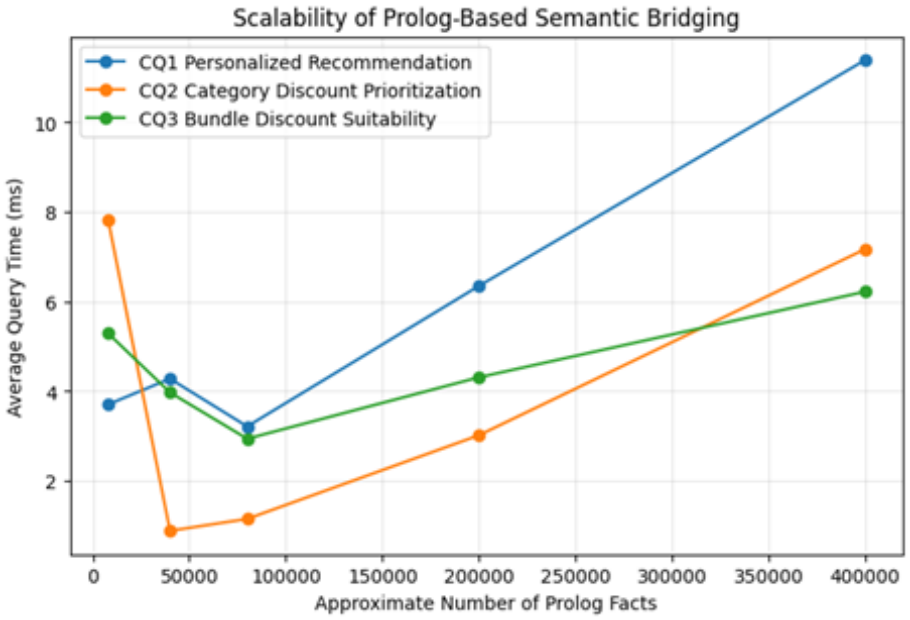


Fig. 2. growth of inference time across the three competency questions as the fact base increases.

As demonstrated in the evaluation, this alignment yields results that are strategically meaningful and semantically coherent, explainable, and computationally stable as the fact base grows.

Future work can advance this research in several concrete methodological directions. A first important line concerns evaluating the proposed framework on more heterogeneous organizational evidence, extending the current onboarding process beyond structured transactional datasets to include documents, semi-structured records, and other distributed business artifacts. This will allow a broader assessment of how robustly the Semantic Onboarding stage can construct unified formal evidence from semantically diverse sources. A second direction involves investigating the simultaneous use of multiple ontologies, rather than a single foundational semantic backbone, in order to analyze issues of ontology interoperability, modular semantic commitments, and cross-ontology alignment during executable knowledge construction. Finally, extending the methodology to more complex business intents, including decision objectives with larger sets of constraints and competing strategic criteria, thus requiring increasingly sophisticated alignment mechanisms between formalized intent and available business evidence.

Supplemental Material

To ensure full reproducibility, all implementation artifacts used in this study, including source code, generated formal models, evaluation datasets, and scalability experiment analysis, are available in [1], enabling the complete replication of both qualitative and quantitative results.

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Declaration

While preparing this work, the authors used Generative AI to assist with grammar, spelling, and text translation. The authors reviewed and edited the content as needed and assume full responsibility for the publication's content.

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[1] https://github.com/aseabramedeiros/semantic_bridging/

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